

A glowing yellow skeleton of a mammoth is positioned on the right side of the image, facing right. The skeleton is complete, showing the skull with long, curved tusks, the ribcage, and the legs. The background is solid black, making the yellow skeleton stand out prominently.

# The Sixth Extinction

## An Unnatural History

Based on the work of Elizabeth Kolbert

*"If there is danger in the human trajectory, it is not so much in the survival of our own species as in the fulfillment of the ultimate irony of organic evolution: that in the instant of achieving self-understanding through the mind of man, life has doomed its most beautiful creations." — E. O. Wilson*



# The Mystery of the Thousand Frog Stream

**THE VICTIM:** Panamanian golden frog (*Atelopus zeteki*).  
**Distinctive features:** Taxicab yellow, toxic skin.  
**Status:** Local symbol of good luck.

**THE SCENE:** El Valle de Antón, Panama. A volcanic crater town where frogs were once so numerous a local creek was named "Thousand Frog Stream."

**THE EVENT:** Early 2000s. A sudden, unexplained vanishing. Populations disappeared from towns and pristine rainforests alike.

**THE RESPONSE:** The El Valle Amphibian Conservation Center (EVACC). A facility functioning as a modern-day Ark, where biologists raced to salvage the last few dozen animals in disinfected tanks.



# The Weedy Species

## The Killer

*Batrachochytrium dendrobatidis* (Bd), or “chytrid.” A fungus that interferes with a frog’s ability to take up electrolytes through its skin, effectively causing a heart attack.

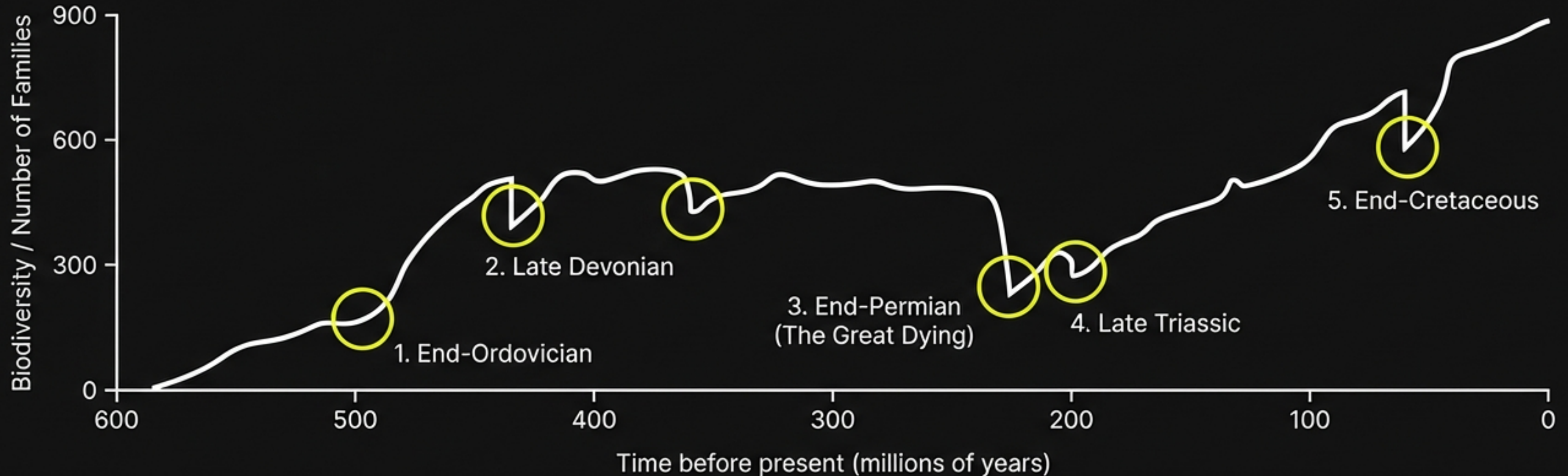
## The Vector

The fungus did not move itself. It hitched a ride on African clawed frogs (exported for pregnancy tests) and North American bullfrogs (exported for food).

## The Implication

Without humans, a frog carrying Bd could never travel from Africa to Australia. One weedy species—us—has unwittingly achieved the ability to directly affect its own fate and that of most other species.

# Background Noise vs. The Crash



## **BACKGROUND EXTINCTION:**

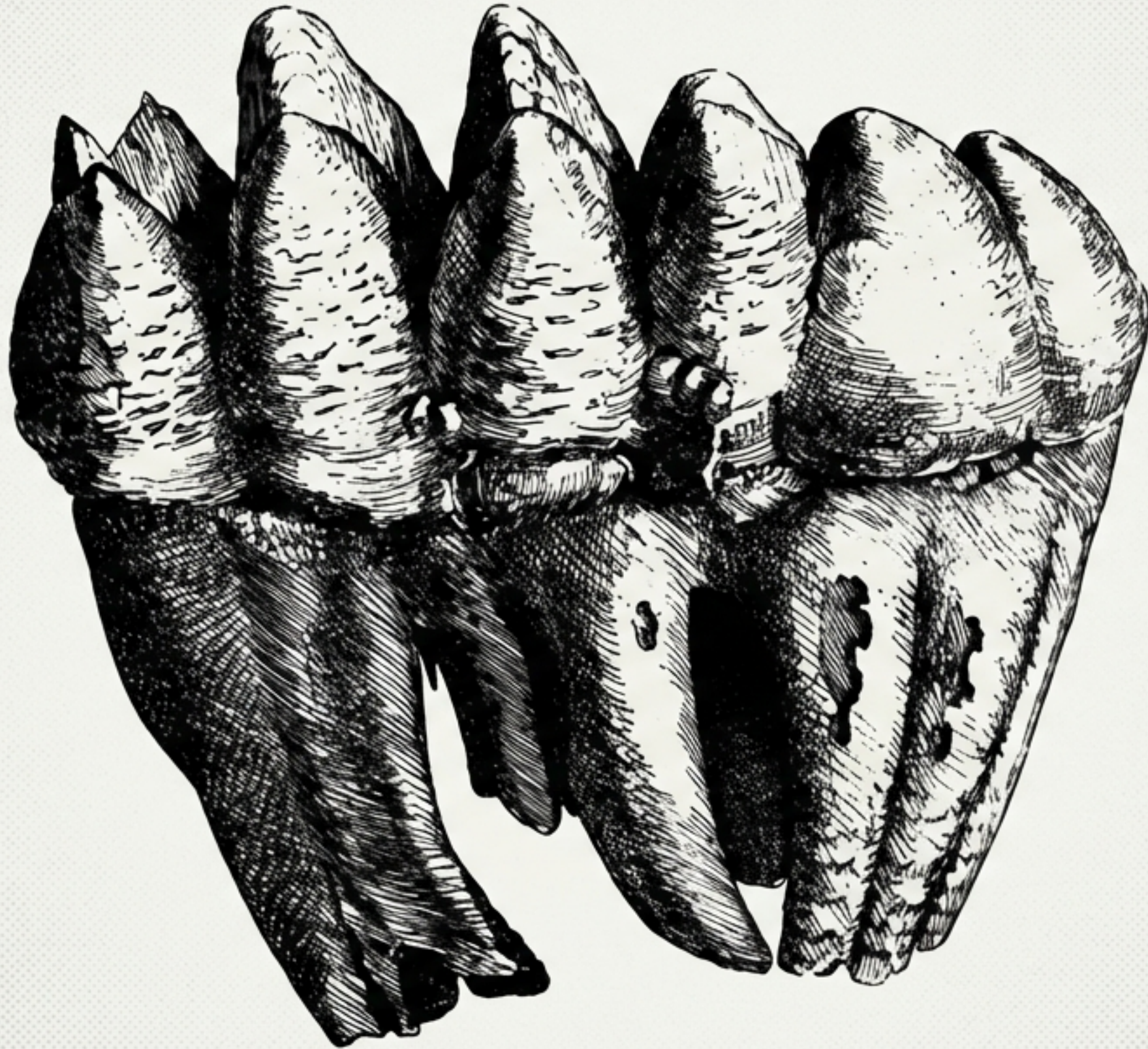
The standard rate of loss. For mammals, roughly one species disappears every 700 years.

## **MASS EXTINCTION:**

Rare, panic-inducing events that eliminate a significant proportion of the world's biota in a geologically insignificant amount of time.

## **CURRENT STATUS:**

The extinction rate for amphibians is estimated to be 45,000 times higher than the background rate. We are witnessing a level of biological annihilation seen only five times in half a billion years.



# The Great Chain of Being

## The Dogma

In the Enlightenment, it was believed no species could die out. The "Great Chain of Being" was considered unbreakable.

## The Anomaly

1739: French troops in the Ohio River valley find giant bones. The femurs looked like elephants, but the teeth were sharp and cusped—like a carnivore's.

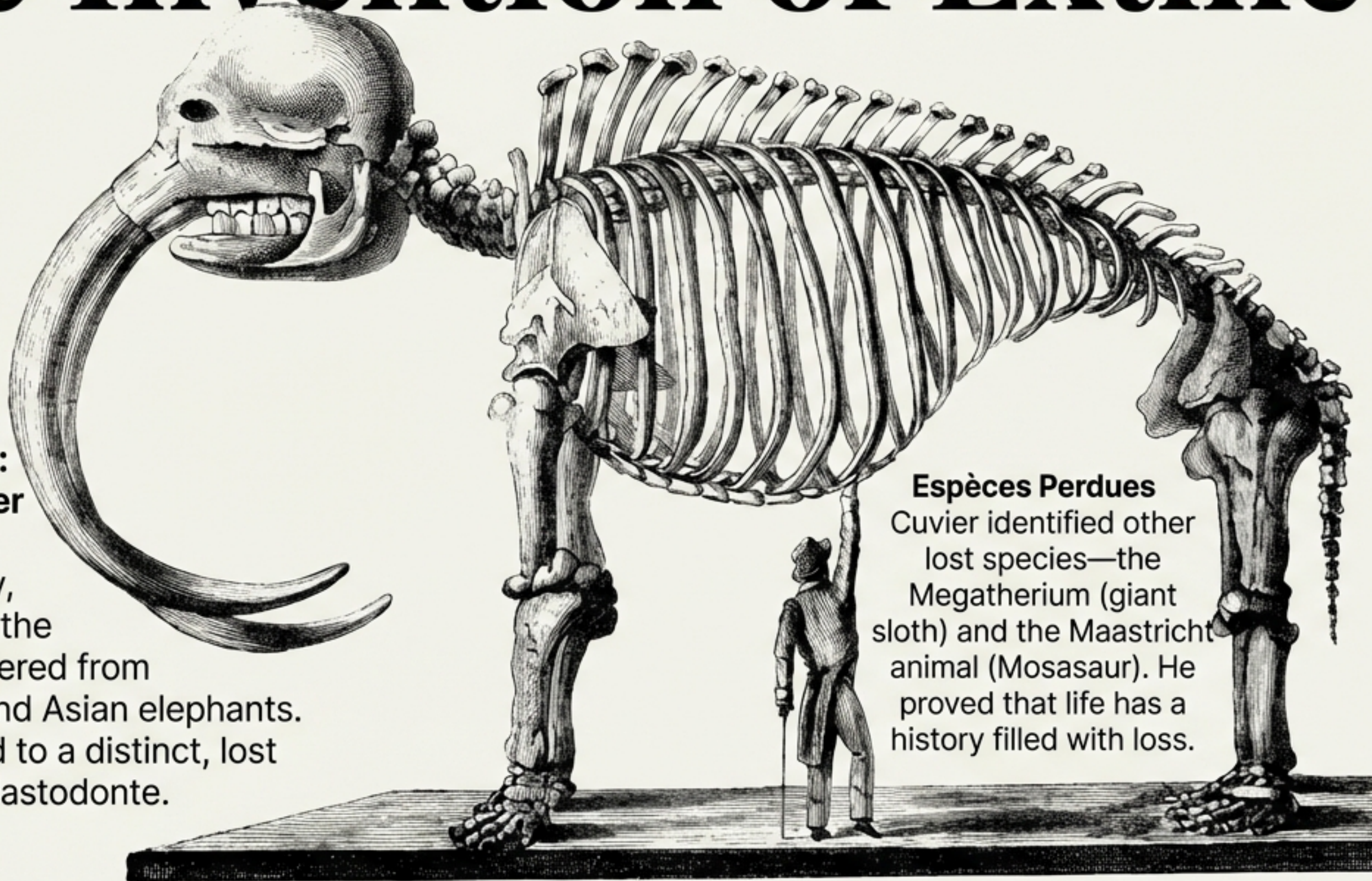
## The Denial

Thomas Jefferson argued that the "American Incognitum" (Mastodon) must still be roaming the unexplored West. He wrote: "Such is the economy of nature, that no instance can be produced of her having permitted any one race of her animals to become extinct."

# The Invention of Extinction

## The Detective: Georges Cuvier (1796)

Using anatomy,  
Cuvier proved the  
Ohio teeth differed from  
both African and Asian elephants.  
They belonged to a distinct, lost  
species: the Mastodonte.



## Espèces Perdues

Cuvier identified other  
lost species—the  
Megatherium (giant  
sloth) and the Maastricht  
animal (Mosasaur). He  
proved that life has a  
history filled with loss.

**Conclusion:** “All these facts... seem to me to prove the existence of a world previous to ours.”

# Revolutions on the Surface of the Earth

## The Theory: Catastrophism

Cuvier argued that the “thread of operations is broken.” Nature has changed course, and agents at work today (rain, erosion) could not have produced the carnage seen in the fossil record.

## The Mechanism

Sudden, violent “catastrophes” or “revolutions”—irruptions of the sea, rapid cooling—wiped out entire faunas.

## The Legacy

Cuvier established that the earth had been dominated by reptiles long before mammals. He saw a direction to history—a succession of lost worlds.



# The Uniformitarian Lull

## The Challenger: Charles Lyell

Thesis: **"The present is the key to the past."**

Geological change is slow, gradual, and constant. There are **no catastrophes, only time.**



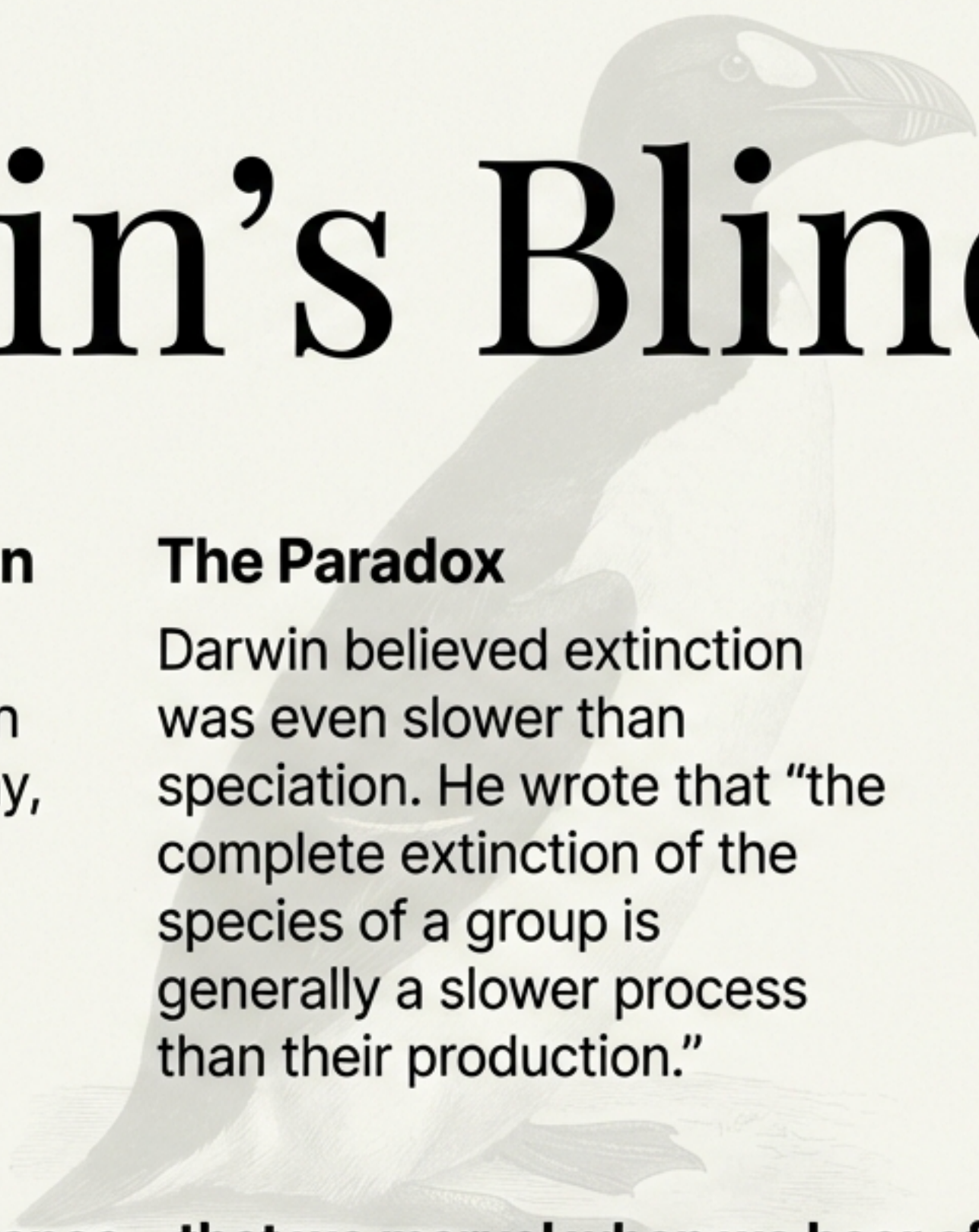
## View on Extinction

Lyell argued that species die out one by one, slowly, like "background noise." He mocked catastrophists for inventing disasters to explain gaps in the record.

## Impact

This view became scientific dogma, deeply influencing a young Charles Darwin.

# Darwin's Blind Spot



## **Descent with Modification**

Darwin “out-Lyelled Lyell,” arguing that natural selection is a ‘daily and hourly’ scrutiny, silently working over vast epochs.

## **The Paradox**

Darwin believed extinction was even slower than speciation. He wrote that “the complete extinction of the species of a group is generally a slower process than their production.”

## **The Reality**

While Darwin wrote that extinction was “slow extermination,” the Great Auk was being slaughtered slaughtered into oblivion in the North Atlantic in real-time.

**“So profound is our ignorance... that we marvel when we hear of the extinction of an organic being; and as we do not see the cause, we invoke cataclysms to desolate the world!”**



# The Original Penguin

## The Species

A flightless bird of the North Atlantic, numbering in the millions. The original “penguins” (from Latin *pinguis*, meaning fat).

## The Decline

By the late 1700s, feather collectors had devastated the colonies.

## The Exploitation

Sailors used them for fresh meat, bait, and fuel. On Funk Island, birds were thrown into fires to cook other birds.

## The Lesson

The Auk didn't fade away due to “natural selection” or competition. It was dismantled, feather by feather, by human industry.



# June 1844: The End of a World

## The Last Stand

After a volcanic eruption destroyed their primary home, the last colony moved to this rock off Iceland.

## The Event

On a morning in June 1844, three Icelanders rowed to Eldey.

## The Act

They found the last pair of Great Auks. Sigurður Iselsson and Jón Brandsson strangled the adults. Ketil Ketilsson smashed the last egg with his boot.

## The Insight

Extinction isn't always a long, slow twilight. It can be a sudden, violent termination. The Auk was gone before Darwin published *Origin of Species*.

# The Message in the Clay

**The Location:** Gubbio, Italy

## The Anomaly

Walter Alvarez found a strange sequence:

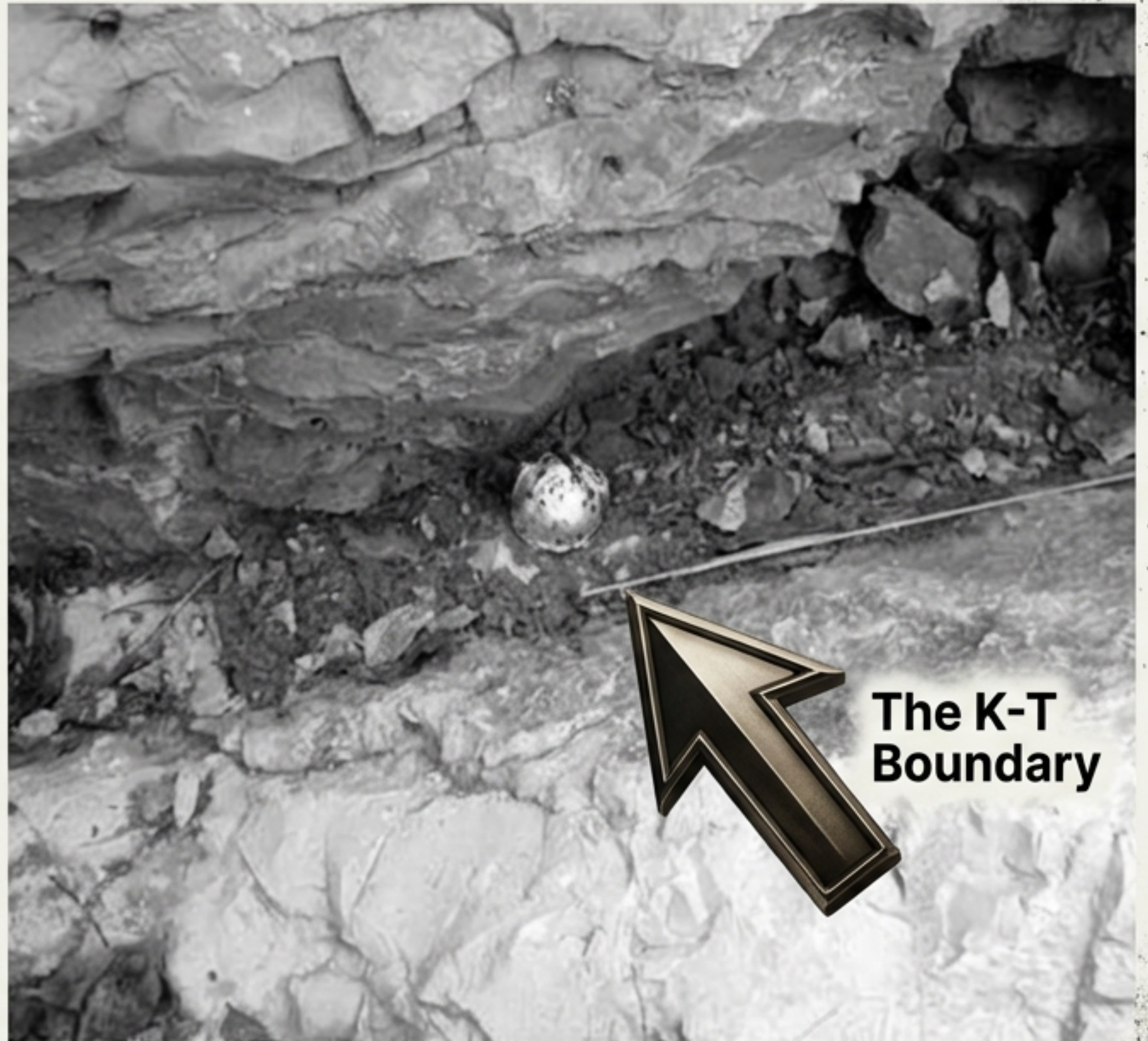
1. Limestone full of diverse, large foraminifera (Cretaceous).
2. A thin layer of clay with no fossils.
3. Limestone with only tiny, primitive foraminifera (Tertiary).

## The Question

How much time did that half-inch of clay represent?

## The Experiment

Luis Alvarez suggested testing for Iridium, a rare element common in meteorites, to clock the deposition.



# The Wild Idea

## The Results

Iridium levels in the clay were astronomical. Confirmed in samples from Denmark and New Zealand.

## The Hypothesis (1980)

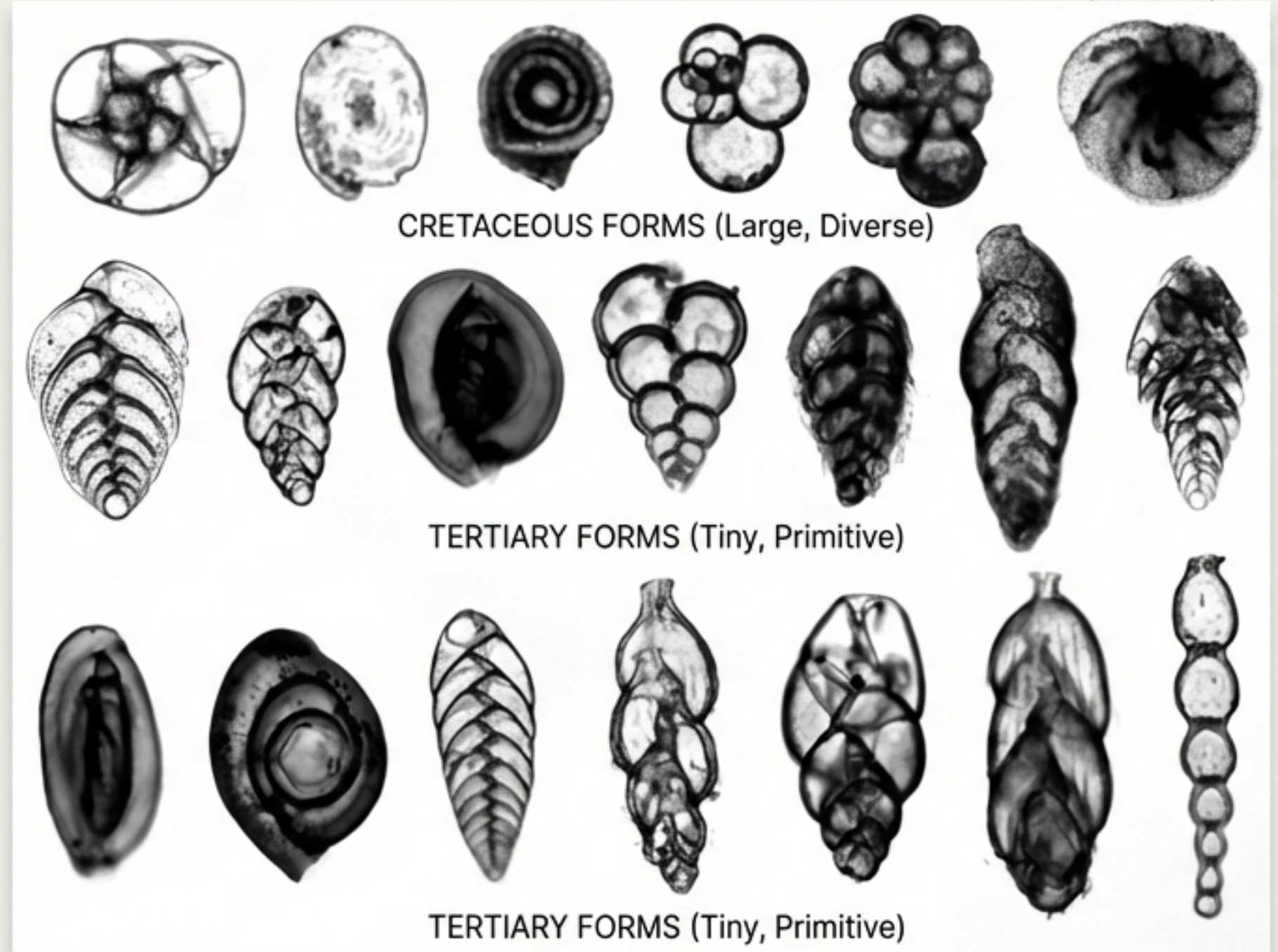
The Alvarezes proposed that an asteroid six miles wide had struck the earth 65 million years ago.

## The Backlash

Paleontologists, trained in Lyell's gradualism, called it "codswallop," arguing extinction was slow and unrelated to "unseen bolides."

## The Proof

A decade later, the "Crater of Doom" (Chicxulub) was identified off the Yucatán Peninsula.



Microfossil evidence showing the dramatic biological change across the K-T boundary.

# The Day the World Ended

## The Impact

100 million megatons of TNT.

## The Aftermath

Debris broiled the planet's atmosphere. Forests were incinerated. The "Fern Spike" replaced complex plants. The oceans became a dead "Strangelove ocean."

## The Casualties

75% of all species. Non-avian dinosaurs, pterosaurs, mosasaurs, and plesiosaurs: 100% extinct.

## The Insight

The "rules of the survival game" changed instantly. Adaptation to the old world meant nothing in the new one.



# The Luck of the Ammonites

## The Victim: Ammonites

Successful for 300 million years.  
They built elaborate, beautiful shells.

## The Survivor: Nautiluses

Similar cephalopods that survived  
the impact.

## The Difference

Ammonites produced tiny, planktonic  
eggs that floated at the surface  
(where conditions were toxic).

Nautiluses laid large eggs in deep  
water.

## The Lesson

Ammonites did nothing wrong.  
Their reproductive strategy  
was perfect for millions of  
years. Then, in a split second,  
it became fatal.

**"The reason we are here is  
because of dinosaurian  
misfortune, not mammalian  
virtue."**



# The Signor-Lipps Effect

## The Illusion

The fossil record is imperfect. Rare species (large circles) seem to disappear before common species (small dots), even if they all died at the same time.

## The Result

Sudden catastrophic boundaries get “smeared out” in the rock record, looking like gradual declines.

## The Takeaway

Darwin’s “much slow extermination” was an optical illusion. Catastrophe hides in plain sight.



# The Tipping Point

## The Graptolites

Colonial creatures that wrote on the rocks of the Ordovician (444 million years ago).

## Dob's Linn, Scotland

A site where rock layers shift suddenly from black (full of life) to gray (barren).

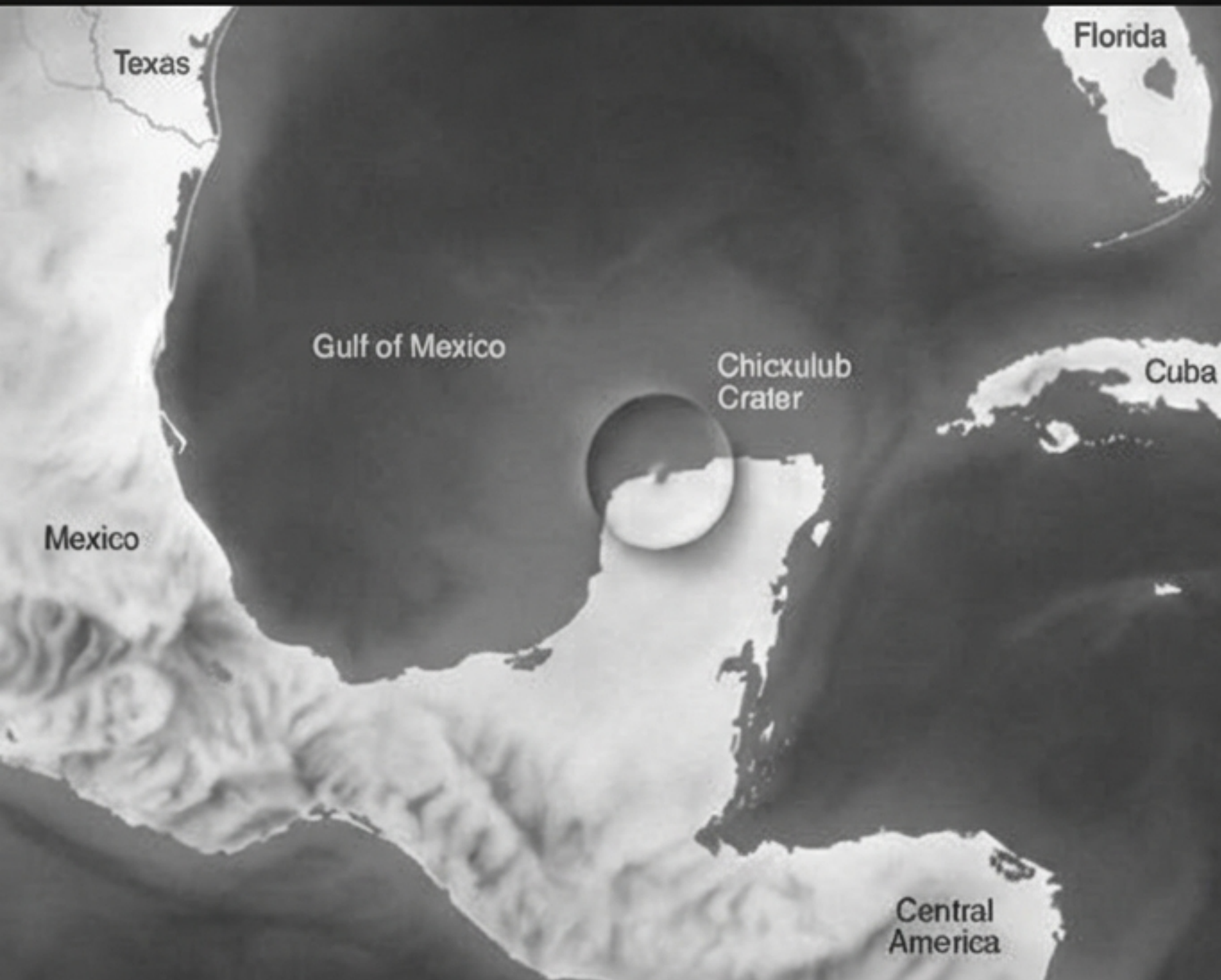
## The Pattern

A sudden shift from a habitable seafloor to an uninhabitable one. "Long periods of boredom interrupted occasionally by panic."

We are currently entering a period of panic.



# The Old Catastrophe



# The New Catastrophe



## Neocatastrophism

The realization that earth's history is driven by rare, devastating events.

## The Sixth Extinction

We are currently causing the Sixth Extinction.

## The Mechanism

Like the asteroid, we are radically altering the atmosphere (climate change) and the ocean (acidification).

## The Rate

We are moving species around the globe (creating a 'New Pangaea') and destroying habitats at a rate comparable to the bolide impact.

# The Ultimate Irony

One weedy species has unwittingly achieved the ability to directly affect its own fate and that of most of the other species on this planet.

## The Difference

The asteroid didn't know it was destroying the dinosaurs. We do.



*“Centuries of centuries and only in the present do things happen.” — Jorge Luis Borges*